APT: Adaptive Pruning and Tuning Pretrained Language Models for Efficient Training and Inference

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Code https://github.com/ROIM1998/APT

- Goal: improve training and inference efficiency of LMs with minimal accuracy loss.
- Key: adaptive pruning (efficiency) + adaptive tuning (accuracy).

Figure 1 placeholder: APT adapter prunes dims/heads and grows ranks (r_apt)

×8	98%	86.4%	−70%
Faster pruning (RoBERTa/T5) vs LoRA+Prune	Task perf. with 60% pruned (small LMs)	LLaMA perf. with 70% params kept	Peak training memory (large LMs)

Introduction

- Scaling LMs improves accuracy but raises training/inference costs (e.g., LLaMA-13B: ~100GB FT, ~30GB INF @fp16).
- Parameter-Efficient Fine-Tuning (PEFT) reduces training memory but not inference cost (model size unchanged).
- Structured pruning speeds inference but increases training time/memory and hurts convergence.
- Challenge: jointly gain training + inference efficiency without large accuracy loss.

Figure placeholder: Problem landscape — Parameter-Efficient Fine-Tuning (PEFT) vs Pruning vs APT

APT: Core Idea

- Early pruning: discard task-irrelevant parameter blocks to cut compute/memory.
- Adaptive tuning: dynamically add salient adapter ranks to recover/boost performance.
- Lightweight salience scoring and efficient search enable low-cost adaptation.
- Self-distillation with shared weights improves accuracy with minimal overhead.

Figure 2 placeholder: Workflow — compute salience → prune blocks → grow ranks → distill

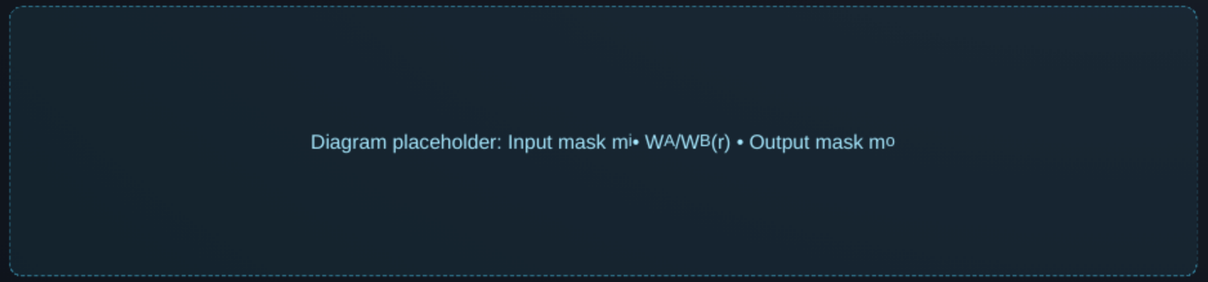
Problem Formulation

- Objective: minimize task loss with target sparsity γ_T after T steps.
- Per step constraints: sparsity $\geq \gamma_t$, tuning params $\leq \Delta_t$.
- Controls: pruning masks M_t and adapter ranks R_t.
- Trade-offs: higher γ → faster inference; larger Δ → faster convergence but more train memory.

We dynamically adjust $C(\Theta_t, M_t)$ and $\delta(\Theta_t, M_t, R_t)$ during fine-tuning.

APT Adapter

- LoRA (Low-Rank Adaptation)-based module with binary masks m_i, m_o and dynamic rank r_{apt}.
- Prunes: hidden dims (inputs), heads in MHA (Multi-Head Attention)/FFN (Feed-Forward Network) neurons (outputs).
- Placement: Q/V in MHA; FFN for smaller models.
- Ranks grow over training; masks prune blocks for efficiency.



Adaptive Pruning (AP)

Salience Scoring

- Outlier-aware: uses activation–gradient magnitudes.
- Batch-summed activations/gradients to save memory.
- Kurtosis combines to preserve outlier-heavy blocks.

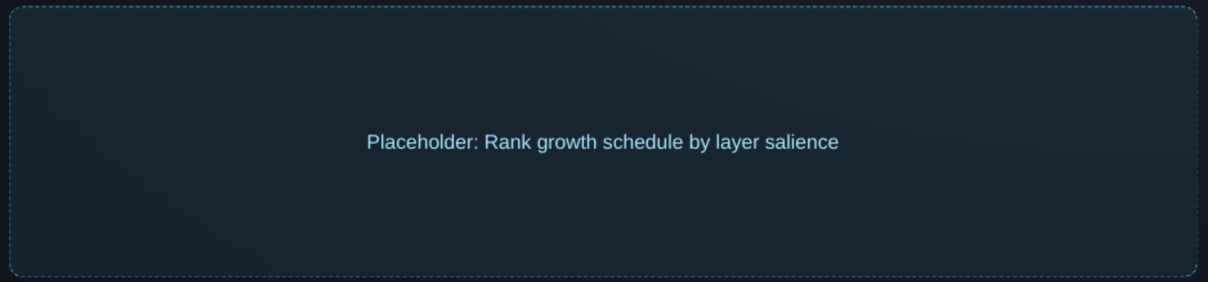
Efficient Block Search

- Score blocks by salience per parameter.
- Binary search threshold to satisfy target sparsity.
- Prune heads, FFN neurons, model dims jointly.

Pruning starts early ($t \ll T$) to cut training time/memory while maintaining stability.

Adaptive Tuning (AT)

- Compute adapter importance $I(H_{apt})$ from parameter salience.
- Increase ranks for top-half salient adapters within budget Δ_t.
- Stable growth: concat Gaussian init on W_A, zeros on W_B (LoRA-style).
- Speeds convergence; recovers pruned performance efficiently.



Efficient Self-Knowledge Distillation (DS)

- Teacher = duplicated tuning layers; shares frozen weights with student.
- Reduces memory and time vs separate teacher models.
- Objective blends supervised FT and dynamic layer-wise distillation.
- Layer mapping with block-wise sampling; transformation initialized as identity.

μ linearly increases from $0 \rightarrow 1$ during training to balance fitting and distillation.

Related Work: Parameter-Efficient Fine-Tuning (PEFT)

What is PEFT?

- Parameter-Efficient Fine-Tuning (PEFT) updates a small subset or additive modules to adapt LMs with low training memory, leaving most weights frozen; inference model size typically unchanged.

Parameter-Efficient Fine-Tuning (PEFT)

- Selective, additive (adapters/LoRA), and dynamic (AdaLoRA).
- Typically training-efficient; inference unchanged.

Compression

- Quantization: memory reduction; speed requires hw support.
- Pruning: unstructured (sparse hw), structured (heads/neurons/dims) for general speedups.

Combining PEFT + Compression

- Prior combos (SPA, CPET, PST, LRP) suffer accuracy drops.
- APT uses salience-driven pruning + dynamic tuning to mitigate losses.

Results

- Small LMs (RoBERTa/T5, 60% sparsity): retain up to 98–99% FT accuracy with up to ×8.4 faster pruning; inference up to ~2.4× faster with ~78–82% memory.
- LLaMA-2 7B (70% params kept): 86.4% average performance with ~75.8% training and ~67.2% inference memory vs LoRA; comparable step time.
- APT improves both training and inference efficiency versus LoRA+Prune, CoFi-based baselines, and LLMPruner at matched sparsities.

Conclusion

- APT couples adaptive pruning with adaptive tuning to jointly optimize training and inference.
- See Results slide for consolidated empirical outcomes; APT maintains strong accuracy under significant parameter reduction.
- Enables resource-friendly fine-tuning across small and large LMs.

Future: broader PEFT integration and improved salience/tuning strategies.

References (Selected)

- Devlin et al., 2019; Liu et al., 2019; Raffel et al., 2020; Kaplan et al., 2020.
- Hu et al., 2022 (LoRA); Zhang et al., 2023b (AdaLoRA); Pfeiffer et al., 2021.
- Xia et al., 2022 (CoFi); Kwon et al., 2022 (Mask Tuning); Lagunas et al., 2021.
- Dettmers et al., 2022; Frantar et al., 2023; Lin et al., 2023 (quantization).
- Ma et al., 2023 (LLMPruner); Zhang et al., 2023a (LRP); Li et al., 2022 (PST).
- Haidar et al., 2022; Hinton et al., 2015 (distillation); Shen et al., 2022.

See paper for full references and appendices.

Appendix: Ablation Highlights

- Adaptive Tuning (AT) restores accuracy and speeds convergence (~16% faster vs static); Adaptive Pruning (AP) crucial for LLaMA memory; removing kurtosis drops LLaMA avg from 50.0→38.1; Distillation (DS) improves accuracy with modest overhead.

One-Sentence Takeaway

APT adaptively prunes and tunes during fine-tuning to deliver up to ×8 faster training with ~70% lower memory while preserving strong task performance (see Results slide).

Appendix: Training Setup Details

- Single A100 GPU; inference batch sizes: 128 (small), 32 (LLaMA 7B), 4 (LLaMA 13B).
- Time-to-Accuracy (TTA) measured at 97% of Full Fine-Tuning (FT) accuracy; efficiency metrics normalized to FT (small) or LoRA (Low-Rank Adaptation) (LLaMA).
- LLaMA trained on Alpaca; GLUE/SQuAD/CNN-DM per paper settings.

Appendix: APT Adapter Placement

- Inserted into Q/V projections of MHA (Multi-Head Attention) for all backbones.
- FFN (Feed-Forward Network) adapters additionally used for smaller LMs (RoBERTa/T5) to speed convergence.
- Masks m_i prune hidden dims; m_o prune heads/neurons.

Appendix: Salience Scoring Details

- Activation–gradient product used due to frozen-weight gradients in Parameter-Efficient Fine-Tuning (PEFT).
- Batch-sum compression before product lowers memory overhead.
- Kurtosis augments block scores to preserve outlier-heavy structures.

Appendix: Block Search Procedure

- Sort blocks by salience per parameter to estimate benefit per cost.
- Binary search threshold to meet target sparsity γ_t.
- Jointly reduce heads, FFN neurons, and hidden dims.

Appendix: Adaptive Tuning Mechanics

- Adapter importance $I(H_{apt})$ = sum of salience over W_B parameters.
- Increase ranks for top-half salient adapters proportionally to Δ growth.
- LoRA (Low-Rank Adaptation)-style init keeps pre-growth function unchanged.

Appendix: Self-Distillation Objective

- Shared frozen weights; duplicated tuning layers as teacher.
- μ schedules $0 \rightarrow 1$ to balance L_{ft} and $L_{distill}$.
- Teacher layers block-wise sampled; mapping to closest non-pruned student layer.

Appendix: Additional Ablations

- RoBERTa: w/o Adaptive Tuning (AT) ≈ LoRA+Prune accuracy; AT improves convergence by ~16%.
- LLaMA: removing kurtosis in salience drops avg from 50.0 to 38.1 (30% sparsity).
- Removing Distillation (DS) saves time/mem but reduces accuracy (~1.35 avg).

Appendix: Efficiency Metrics Normalization

- Training speed reported as time per step (LLaMA) or Time-to-Accuracy (TTA) (small models).
- Inference speed via throughput; memory via peak allocated GPU memory.
- RoBERTa/T5 normalized to Full Fine-Tuning (FT); LLaMA to LoRA (Low-Rank Adaptation) baseline.

Appendix: Discussion & Limitations

- Pruned LLaMA gap larger due to distillation-free setup; future: memory-efficient distillation.
- Hardware-friendly shapes (e.g., multiples of 8/16) can yield better real speedups.
- Dynamic parameter changes may introduce instability; optimizer resets help.
- Explore richer adapters (prefix, parallel) while preserving inference efficiency.

Appendix: Acronyms and Definitions

- PEFT — Parameter-Efficient Fine-Tuning: adapts LMs by updating a small set/additive modules; inference size unchanged.
- AP — Adaptive Pruning: early, salience-based removal of heads/neurons/dims to cut compute and memory.
- AT — Adaptive Tuning: dynamically grows adapter ranks in salient layers within a budget.
- DS — (Self-)Knowledge Distillation: teacher–student training with shared frozen weights to recover accuracy.
- LoRA — Low-Rank Adaptation: additive low-rank matrices enabling memory-efficient fine-tuning.
- FFN — Feed-Forward Network: transformer MLP sublayer whose neurons can be pruned.
- MHA — Multi-Head Attention: transformer attention with multiple heads that can be pruned.